

Prove monotonicity of First Fit Decreasing packing for large bin capacity scaling

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Editors' note. The resolution is entirely driven by a zero-cost chore. It gives no progress on the natural repaired problem in which every chore has strictly positive cost, nor does it refute the paper's weaker notion of threshold monotonicity.

Abstract

We give a complete negative solution to the stated strong monotonicity problem. For every integer $n \geq 2$ and every real number $\alpha \geq 8/7$, we construct a finite nonnegative additive instance for which first-fit decreasing with threshold $\alpha \text{MMS}(M, v, n)$ allocates every item, while an n -bundle FFV collection at the same threshold is incomplete. In fact, the same construction works for every $\alpha \geq 1$. Thus no gap remains for the problem as stated. The construction uses an item of value zero, which is permitted by the nonnegativity hypothesis and is indistinguishable from zero padding in the lexicographic comparison.

1 The open problem

This paper addresses an open problem posed by Xin Huang, Erel Segal-Halevi in *A Reduction from Chores Allocation to Job Scheduling* [1] (Economics and Computation (EC), 2023), Section 8 (Monotonicity in special cases), page 24; reiterated in Section 9 (Future Work), page 25.. The website catalogue entry is problem 8 (W4383533394_p0) of the [Open Problems in Operations Research](#) collection.

For every integer $n \geq 2$ and every real $\alpha \geq \frac{8}{7}$, determine whether FFD is monotone with respect to (α, n) (as defined above). Equivalently, decide whether for all instances (M, v) with $\mu = \text{MMS}(M, v, n)$,

$$\text{FFD}(M, v, \alpha\mu, n) \text{ allocates all items} \implies \left(\forall (M, A, v, \alpha\mu) \text{ FFV with } |A| = n, \bigcup_{k=1}^n A_k = M \right).$$

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_030926_630f21f9, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Definitions and the monotonicity assertion

Let M be a finite set of items, and let

$$v: M \longrightarrow \mathbb{R}_{\geq 0}$$

be a nonnegative value function. We extend v additively to subsets:

$$v(S) = \sum_{c \in S} v(c) \quad \text{for every } S \subseteq M.$$

For an integer $n \geq 1$, let $\Pi_n(M)$ denote the set of ordered n -tuples $(P_1, \dots, P_n)$ of pairwise disjoint subsets of M whose union is M ; empty bundles are permitted. Define

$$\text{MMS}(M, v, n) = \min_{(P_1, \dots, P_n) \in \Pi_n(M)} \max_{1 \leq i \leq n} v(P_i).$$

Thus $\text{MMS}(M, v, n)$ is the minimum possible makespan of a complete allocation of M to n bins.

Fix a threshold $\tau \geq 0$. The first-fit decreasing procedure FFD begins with n empty bins. It orders the items in nonincreasing order of value, resolving equal values by an arbitrary fixed tie-breaking rule, and then processes the items in that order. When processing an item c , it scans the bins in the order $1, \dots, n$ and places c in the first bin i for which

$$v(P_i) + v(c) \leq \tau.$$

If no such bin exists, the item is not allocated and the procedure fails. We say that FFD *succeeds* if it allocates every item of M .

We next state the lexicographic convention used in the FFV definition. For a finite subset $S \subseteq M$, list the values of its items in nonincreasing order and pad the resulting list with zeros indefinitely. The resulting sequence is denoted by

$$\lambda(S) = (\lambda_1(S), \lambda_2(S), \dots).$$

Explicitly, if S has item values $s_1 \geq \dots \geq s_m$, then

$$\lambda_j(S) = \begin{cases} s_j, & 1 \leq j \leq m, \\ 0, & j > m. \end{cases}$$

In particular, $\lambda(\emptyset) = (0, 0, 0, \dots)$. These sequences are compared in lexicographic order: $\lambda(S)$ is strictly larger than $\lambda(T)$ if, at the first coordinate at which the two sequences differ, the coordinate of $\lambda(S)$ is larger.

Let $A = (A_1, \dots, A_n)$ be an ordered collection constructed sequentially, and define the residual set before the k -th bundle by

$$R_k = M \setminus \bigcup_{i < k} A_i.$$

A subset $S \subseteq R_k$ is *feasible* at stage k if $v(S) \leq \tau$. A *benchmark* B_k at stage k is any feasible subset of R_k whose padded value sequence is lexicographically maximal among all feasible subsets of R_k ; since M is finite, such a benchmark exists. The collection A is an *FFV collection* at threshold τ if, for every $k = 1, \dots, n$,

$$A_k \subseteq R_k, \quad v(A_k) \leq \tau, \quad \text{and} \quad \lambda(A_k) = \lambda(B_k)$$

for a benchmark B_k ; equivalently, if $\lambda(A_k)$ is lexicographically maximal among the padded value sequences of all feasible subsets of R_k .

The strong monotonicity assertion at parameters (α, n) is the implication

$$\begin{aligned} &\text{FFD succeeds with threshold } \tau = \alpha \text{MMS}(M, v, n) \\ \implies &\bigcup_{k=1}^n A_k = M \quad \text{for every FFV collection } A \text{ at threshold } \tau. \end{aligned}$$

We prove that this implication is false for every pair of parameters requested in the problem.

3 Complete counterexample

Theorem 1. *For every integer $n \geq 2$ and every real number $\alpha \geq 1$, there exist a finite item set M , a nonnegative additive value function v , and an n -bundle FFV collection A such that, for*

$$\tau = \alpha \text{MMS}(M, v, n),$$

first-fit decreasing succeeds at threshold τ , but

$$\bigcup_{k=1}^n A_k \neq M.$$

Consequently, strong monotonicity fails for every $n \geq 2$ and every $\alpha \geq 8/7$.

Proof. Fix an arbitrary integer $n \geq 2$ and an arbitrary real number $\alpha \geq 1$. Let

$$M = \{a, z\}, \quad v(a) = 1, \quad v(z) = 0.$$

The value function is nonnegative, as required.

We first compute $\text{MMS}(M, v, n)$. Consider any complete partition $(P_1, \dots, P_n) \in \Pi_n(M)$. Because the bundles cover M , there is an index j such that $a \in P_j$, and nonnegativity together with additivity gives

$$v(P_j) \geq v(a) = 1.$$

Therefore $\max_{1 \leq i \leq n} v(P_i) \geq 1$ for every complete partition, and hence

$$\text{MMS}(M, v, n) \geq 1.$$

For the reverse inequality, consider the partition

$$P_1 = \{a, z\}, \quad P_2 = \dots = P_n = \emptyset,$$

which is admissible because empty bundles are permitted. Its bundle values are $1, 0, \dots, 0$, so $\max_{1 \leq i \leq n} v(P_i) = 1$, and thus

$$\text{MMS}(M, v, n) \leq 1.$$

Combining the two inequalities yields $\text{MMS}(M, v, n) = 1$, so the prescribed threshold is

$$\tau = \alpha \text{MMS}(M, v, n) = \alpha.$$

We now run first-fit decreasing. Since $v(a) = 1 > 0 = v(z)$, the item a is processed before z , independently of the fixed rule used to break ties between equal-valued items. Because $\alpha \geq 1$, we have $v(a) = 1 \leq \tau$, so a is placed in the first bin, whose load is then 1. When z is processed, adding it does not change the load:

$$1 + v(z) = 1 + 0 = 1 \leq \tau.$$

Consequently, z is also placed in the first bin. Thus FFD allocates every item of M .

It remains to construct an incomplete FFV collection. Define

$$A_1 = \{a\}, \quad A_2 = \cdots = A_n = \emptyset.$$

We verify the FFV condition separately at the first stage and at all later stages.

At stage $k = 1$, the residual set is $R_1 = M = \{a, z\}$. Since $v(M) = 1 \leq \tau$, every subset of M is feasible. The padded value sequences of these subsets are

$$\begin{aligned} \lambda(\emptyset) &= (0, 0, 0, \dots), & \lambda(\{z\}) &= (0, 0, 0, \dots), \\ \lambda(\{a\}) &= (1, 0, 0, \dots), & \lambda(\{a, z\}) &= (1, 0, 0, \dots). \end{aligned}$$

The lexicographically maximal feasible sequence is therefore $(1, 0, 0, \dots)$, and both possible benchmark bundles containing a , namely $\{a\}$ and $\{a, z\}$, have exactly this sequence. Since $A_1 = \{a\}$, we have

$$\lambda(A_1) = (1, 0, 0, \dots),$$

so A_1 satisfies the FFV condition at stage 1, regardless of which maximizing benchmark is selected.

Now let k be any integer with $2 \leq k \leq n$. Because $A_1 = \{a\}$ and $A_2 = \cdots = A_{k-1} = \emptyset$, the residual set is

$$R_k = M \setminus \bigcup_{i < k} A_i = \{z\}.$$

The only subsets of R_k are $\emptyset$ and $\{z\}$. Both are feasible because they have value zero, and their padded value sequences coincide:

$$\lambda(\emptyset) = \lambda(\{z\}) = (0, 0, 0, \dots).$$

Hence $(0, 0, 0, \dots)$ is the lexicographically maximal feasible sequence at stage k . Since $A_k = \emptyset$, it follows that $\lambda(A_k) = (0, 0, 0, \dots)$, so A_k satisfies the FFV condition. This argument applies to every $k = 2, \dots, n$. Therefore $A = (A_1, \dots, A_n)$ is an FFV collection at threshold τ .

Nevertheless,

$$\bigcup_{k=1}^n A_k = \{a\} \neq \{a, z\} = M.$$

Thus the FFV collection is incomplete even though first-fit decreasing succeeds at the same threshold. This disproves strong monotonicity for every $n \geq 2$ and every $\alpha \geq 1$, and therefore in particular for every $n \geq 2$ and every $\alpha \geq 8/7$. $\square$

4 Scope of the conclusion

The theorem gives a complete solution, rather than partial progress, for the stated problem with nonnegative item values. The proof covers the boundary $\alpha = 8/7$, the boundary $n = 2$, every larger integer n , and arbitrarily large values of α ; indeed, only the weaker inequality $\alpha \geq 1$ was used.

The counterexample exhibits a genuine failure of completeness: first-fit decreasing allocates all of M , whereas the displayed FFV collection literally leaves the item z outside $\bigcup_{k=1}^n A_k$. The construction is unaffected by benchmark nonuniqueness, since all maximizing benchmarks at each relevant stage have the same padded value sequence. It is also unaffected by the fixed item tie-breaking rule, since $v(a) > v(z)$.

The mechanism depends on the permission to use zero-valued items together with zero padding in the lexicographic comparison: the padded sequences of $\{a\}$ and $\{a, z\}$ coincide, as do those of $\emptyset$ and $\{z\}$. If the hypothesis were changed to require $v(c) > 0$ for every item c , then replacing $v(z) = 0$ by any $\varepsilon > 0$ would give

$$\lambda(\{a, z\}) = (1, \varepsilon, 0, \dots) >_{\text{lex}} (1, 0, 0, \dots) = \lambda(\{a\}),$$

so this particular FFV construction would no longer work. That strictly positive variant is a different problem and is not claimed to be resolved here; it is not a gap in the complete negative solution under the stated nonnegativity hypothesis.

References

- [1] Xin Huang, Erel Segal-Halevi. *A Reduction from Chores Allocation to Job Scheduling*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597676>